# Buy now above the fold

*Rendering the Buy Now call-to-action above the fold on the product detail page, relative to the current below-fold placement, increases conversion_rate among product-page visitors on the demo storefront over the exposure window, with the expected direction being a higher per-user probability of placing at least one order. The treatment is hypothesized to act by reducing the interaction distance between product consideration and purchase initiation; the magnitude of the effect is pre-registered in this brief at the feasibility-bounded relative MDE below and is not asserted as a point prediction.*

AT A GLANCE

**Primary metric is *conversion_rate*; decision is binary against a positive lower bound on conversion lift AND no guardrail breach.**

*Pre-registered before any analysis. Decision is binary against conversion_rate: a 7.0% relative relative lift over a 10.0% baseline requires 29,718 per arm at the standard a / power settings. Guardrails: revenue_per_user, page_load_time, cart_abandonment_rate.*

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SAMPLE REQUIRED (PER ARM)

**29,718**

*to detect a 7.0% relative relative effect at a = 0.05, power = 0.80*

BASELINE
**10.0%**

DURATION
**~7.7 days at 7,735/day accrual**

TOTAL SAMPLE
**59,436**

FEASIBILITY
**VIABLE**

MECHANISM

## Why we think this works

*An above-fold Buy Now call-to-action reduces the steps between product consideration and purchase initiation; the treatment is expected to lift the per-user probability of placing at least one order.*

*Conversion baseline on the demo storefront sits near 10% over the trailing 28 days.*

METRICS

## conversion_rate is the verdict-driving metric.

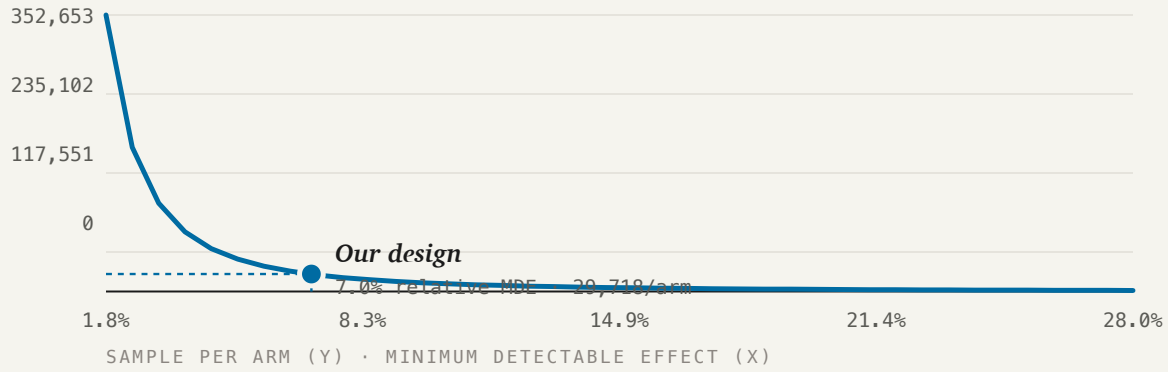
*Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their CI crosses the pre-registered non-inferiority margin in the adverse direction.*

| | | |
|---|---|---|
| PRIMARY | **conversion_rate** | Proportion · baseline 10.0% · MDE 7.0% relative · direction higher is better |
| GUARDRAIL | **revenue_per_user** | Direction higher is better. Decision blocks if 90% CI lies past -1.0% relative. |
| GUARDRAIL | **page_load_time** | Direction lower is better. Decision blocks if 90% CI lies past 5.0% relative. |
| GUARDRAIL | **cart_abandonment_rate** | Direction lower is better. Decision blocks if 90% CI lies past 5.0% relative. |

POWER

## Detecting smaller effects requires much more sample.

*To resolve a 7.0% relative relative effect against a 10.0% baseline, the design requires 29,718 per arm. Tighter detection thresholds climb the curve to the left.*




*Smaller effects require disproportionately more sample. Our design point is feasible inside the available user_id surface.*

ASSIGNMENT + COHORTS

## First-exposure attribution at the user_id level.

*Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before first_exposure_at are not attributed to the experiment.*

| | |
|---|---|
| UNIT | **user_id** |

| | |
|---|---|
| | One row per (user_id, experiment_id) in the assignments table; first_exposure_at gates which events are attributed. |
| ARMS | **50 / 50**<br>control · treatment |
| COHORT | **Users on the demo storefront whose first exposure to the experiment is their first `page_event` row with `event_name = 'product_view'` during the exposure window; assignment fires at that event (`assignment.first_exposure_at = MIN(page_event.event_at WHERE event_name = 'product_view')` per `user_id`), so "all assigned users" is by construction the set of product-page visitors during the window, segment = general.** |

DECISION RULES

## Pre-registered. The analyze verb walks the same tree against the same metrics.

*Ship the treatment if and only if the two-sided 95% confidence interval on the relative lift in conversion_rate excludes zero on the positive side (lower bound strictly greater than zero) AND no guardrail breaches its pre-registered non-inferiority margin. If the primary interval crosses zero, or if any guardrail's confidence interval crosses its non-inferiority margin in the adverse direction, the verdict is do-not-ship regardless of point estimate. The rule is intentionally tight: a treatment whose true effect is below the pre-registered MDE is expected to fail it.*

| | |
|---|---|
| SHIP | Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7 |
| LIFT WITH CAVEAT | Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out |
| NO-SHIP · GUARDRAIL | Any guardrail's 90% CI lies past its harm threshold |
| NO-LIFT | Well-powered, primary CI straddles zero with half-width > 2 × MDE |
| INCONCLUSIVE | Underpowered (n < required) AND primary CI straddles zero |
| INVALID | SRM $\chi^2$ $p < 0.0005$ and no override resolved — investigate assignment |

INTEGRITY LOCK · THREE-PART SEAL

## brief.sealed.yaml binds intent + metric definitions + expected shape.

*Tampering with any of the three components after seal will fail `agentxp.schemas.brief_seal.verify_or_raise`; `agentxp analyze`*

*refuses to open a brief whose seal does not verify (R11). This is the architectural wall, not a guideline.*

| | |
|---|---|
| DESIGN_CHAIN_HASH | 37770e1e22d1d584de8602b37156d00854a1c79702a1cd12e0829393849b1a13 |
| METRIC_SNAPSHOT | 4 metrics hashed |
| CONVERSION_RATE | f775e246e2992c91fa7a04f5d22bb93d2cb9ed5078144479b7792e6cdf9a5fe1 |
| REVENUE_PER_USER | c4742f81560b2836b851f2c17bdd1bd3cb102fdf14a69a04cd9170b3cf9d69ae |
| PAGE_LOAD_TIME | f1914747b5fdf7aac000b3991e59262b5871c3fe85bfc18a82e8bfc7084563e4 |
| CART_ABANDONMENT_RATE | 6498a159e387655f28f9949108bee2f9954e7078afa745f6f0430c2c5eb36db5 |
| EXPECTED_SHAPE | user_id · 50 / 50 |
| SEALED_AT | 2026-06-05T18:43:44.845392Z |
| SEALED_BY | shane@aieval.ai |

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